

Data Source Files Download and Preparation Instructions

Data preparation Jupyter notebook : `student_loan_project_data_preparation.ipynb`

The raw source files used in this project are very large and come from multiple federal data portals (College Scorecard, Federal StudentAid, and ACS microdata). Because these datasets are too big and sparse to store directly with the project, they must be downloaded from the official sources by following the instructions below. After downloading, each source is cleaned, filtered, and aggregated in the Jupyter notebook `student_loan_project_data_preparation.ipynb`.

The cleaning code produces compact state-year CSV files:

- **scorecard_state_year.csv** — state-level financial and sector variables derived from College Scorecard
- **acs_state_year_controls.csv** — demographic and socioeconomic controls constructed from the ACS microdata extract
- **fsa_state_year_annualized.csv** — FAFSA application volumes aggregated across the quarterly state-level files These three cleaned datasets are then merged into a single analysis-ready panel:
- **state_year_joined_2011_2020.csv** This joined file is the only dataset used for further analysis.

Dataset 1: [College Scorecard](#)

- Source: U.S. Department of Education (Public API / CSV)
- Scope: Institution-level data on tuition, median debt, earnings, completion, selectivity, and program mix.
- Use: Adds institutional context (net price, default rates, median earnings).
- Time frame: Annual, 1996–present.
- Join Key: OPEID or cross-walk from NPSAS / IPEDS IDs, Year.

Steps:

1. click the dark green button with "download(.zip,390MB)" (this file size number changes by time) next to "All Data Files", zip file name is `College_Scorecard_Raw_Data_10032025.zip`.
2. merge all the `MERGEDYYYY_YY_PP.csv` files as our combined college scorecard data.

Download the data that appear on the College Scorecard, as well as supporting data on student completion, debt and repayment, earnings, and more.

This data was **last updated April 23, 2025**.

All Data Files

Download (.zip, 390 MB) 

- Institution-level data files for 1996-97 through 2022-23 containing aggregate data for each institution. Includes information on institutional characteristics, enrollment, student aid, costs, and student outcomes.
- Field of study-level data files for the pooled 2014-15, 2015-16 award years through the pooled 2018-19, 2019-20 award years containing data at the credential level and 4-digit CIP code combination for each institution. Includes information on cumulative debt at graduation and earnings one year after graduation.
- Crosswalk files for 2000-01 through 2022-23 that link the Department's OPEID with an IPEDS UNITID for each institution.

Most Recent Institution-Level Data

Download (.zip, 22 MB) 

Most Recent Data by Field of Study

Download (.zip, 13 MB) 

Dataset 2 - Federal StudentAid

Download link: <https://studentaid.gov/data-center/student/application-volume/fafsa-school-state>

Federal StudentAid
AN OFFICE of the U.S. DEPARTMENT of EDUCATION

FAFSA® Form ▾

Grants and Loans ▾

Loan Repayment ▾

Loan Forgiveness ▾

On This Page

FAFSA Data by Demographic Characteristics

FAFSA by Postsecondary School:

FAFSA Data by State

FAFSA Report Definitions

FAFSA Data by State

2025-2026	▾
2024-2025	▾
2023-2024	▾
2022-2023	▾

Steps:

1. Go to page via above link.
2. On the page, find "FAFSA by State"
3. Under each year range, such as "2021-2022", click the blue link such as " 2021 2022 Q7", then xls will be downloaded automatically.
4. Click all the links of the desired years to download the data
5. Then merge into 1 single csv file.

Dataset 3: ACS (American Community Survey)

Download link: <https://usa.ipums.org/usa/acs.shtml>

- Run by the U.S. Census Bureau.
- A large annual survey of U.S. households (about 3.5 million per year). Contains: demographics (age, race, education, migration), housing, income, and employment (occupation, industry, wages, commute, telework).
- Microdata (IPUMS) = person-level records you can analyze directly (instead of just summary tables).
- Data can be tailored and downloaded upon submitting the request.
- Downloaded version cover the variables as below and range(2010-2023)
- Variables as below:

In cart	Variable	Variable Label	Type	Codes	2023 acs	2022 acs	2021 acs	2020 acs	2019 acs	2018 acs	2017 acs	2016 acs	2015 acs	2014 acs	2013 acs	2012 acs	2011 acs	2010 acs
☒	<u>YEAR</u>	Census year <u>[preselected]</u>	H	<u>codes</u>	X	X	X	X	X	X	X	X	X	X	X	X	X	X
☒	<u>HHWT</u>	Household weight <u>[preselected]</u>	H	<u>codes</u>	X	X	X	X	X	X	X	X	X	X	X	X	X	X
☒	<u>GQ</u>	Group quarters status <u>[preselected]</u>	H	<u>codes</u>	X	X	X	X	X	X	X	X	X	X	X	X	X	X
☒	<u>STATEFIP</u>	State (FIPS code)	H	<u>codes</u>	X	X	X	X	X	X	X	X	X	X	X	X	X	X
☒	<u>HHINCOME</u>	Total household income	H	<u>codes</u>	X	X	X	X	X	X	X	X	X	X	X	X	X	X
☒	<u>PERNUM</u>	Person number in sample unit <u>[preselected]</u>	P	<u>codes</u>	X	X	X	X	X	X	X	X	X	X	X	X	X	X
☒	<u>PERWT</u>	Person weight <u>[preselected]</u>	P	<u>codes</u>	X	X	X	X	X	X	X	X	X	X	X	X	X	X
☒	<u>SEX</u>	Sex	P	<u>codes</u>	X	X	X	X	X	X	X	X	X	X	X	X	X	X
☒	<u>AGE</u>	Age	P	<u>codes</u>	X	X	X	X	X	X	X	X	X	X	X	X	X	X
☒	<u>RACE</u>	Race	P	<u>codes</u>	X	X	X	X	X	X	X	X	X	X	X	X	X	X
☒	<u>HISPAN</u>	Hispanic origin	P	<u>codes</u>	X	X	X	X	X	X	X	X	X	X	X	X	X	X
☒	<u>SCHOOL</u>	School attendance	P	<u>codes</u>	X	X	X	X	X	X	X	X	X	X	X	X	X	X
☒	<u>EDUC</u>	Educational attainment	P	<u>codes</u>	X	X	X	X	X	X	X	X	X	X	X	X	X	X
☒	<u>EMPSTAT</u>	Employment status	P	<u>codes</u>	X	X	X	X	X	X	X	X	X	X	X	X	X	X
☒	<u>INCTOT</u>	Total personal income	P	<u>codes</u>	X	X	X	X	X	X	X	X	X	X	X	X	X	X
☒	<u>POVERTY</u>	Poverty status	P	<u>codes</u>	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Steps:

1. register with GT email, and login.
2. On homepage, click "select data" on the top bar.
3. click "select samples", go to its page.
4. check the year box for ACS (1996 – 2023 this is the range I selected. But we only use 2011 to 2020 in the end) and uncheck the content not desired. then click "submit sample selections" button at the bottom of the page.
5. select the variables, click the "plus" sign in front of the desired variable, then it will be added to the cart.
6. The quick way is using the "search" to find the desired variables and add to cart:
'YEAR'HHWT'STATEFIP'GQ'HHINCOME'PERNUM'PERWT'SEX'AGE'RACE'RACED'
HISPAN'HISPAND'SCHOOL'EDUC'EDUCD'EMPSTAT'EMPSTATD'INCTOT'POVERTY'
7. After selecting the wanted variables, click "view cart" button.
8. Review the select vars, then click "create data extract"
9. IPUMS.org will review this extract request shortly.
10. Download the dataset once it's got approved.